

Probabilistic Graphical Models II: Bayesian Network Representation

Statistical Data Mining II

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Outline

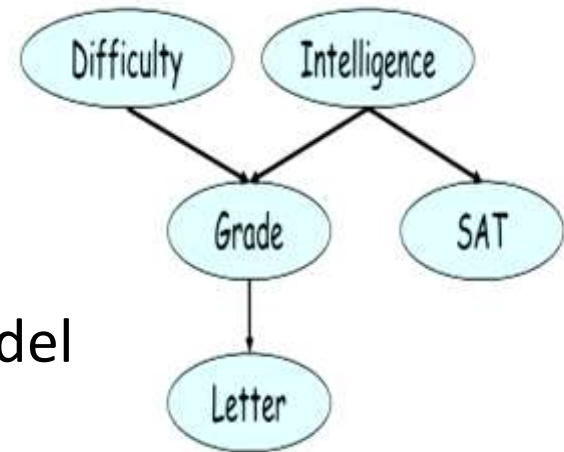
- Flow of Probabilistic Reasoning
- Active trail
- D-separation and Independence
- Naïve Bayes
- Applications
- The learning task
- Special cases
- What do we do with the data?
- Selection of hyper parameters

Bayesian Network

When can X influence Y:

(when we condition on X when does it change out belief in Y).

- $X \rightarrow Y$ - yes, straight downward path
- $X \leftarrow Y$ - yes, evidential reasoning
- $X \rightarrow W \rightarrow Y$ - yes, indirect
- $X \leftarrow W \leftarrow Y$ - yes, reverse
- $X \leftarrow W \rightarrow Y$ - yes,
- $X \rightarrow W \leftarrow Y$ - no V-structure/collider model



Bayesian Network

A trail $X_1 - X_2 - \square - X_k$ is **active** if it has no v-structures:

$$X_{i-1} \rightarrow X_i \leftarrow X_{i+1}.$$



General edge type up/down



A block in the trail

Therefore, information can flow freely through the network, in the “active” sense, unless a v-structure arises.

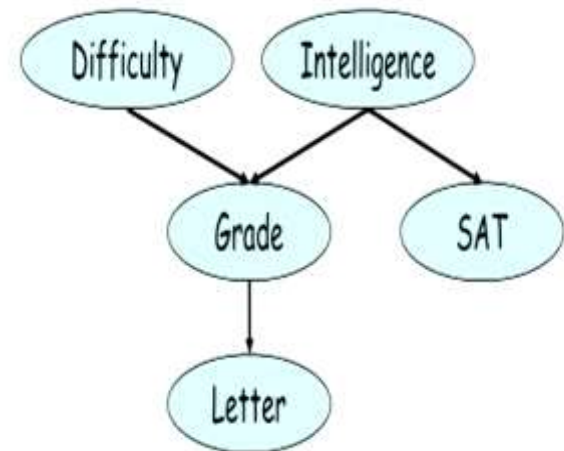
**Lets think about additional evidence*

Bayesian Network

When can X influence Y given evidence about Z:

$$X \rightarrow Y$$

$$X \leftarrow Y$$



Bayesian Network

When can X influence Y given evidence about Z:

$$X \rightarrow Y$$

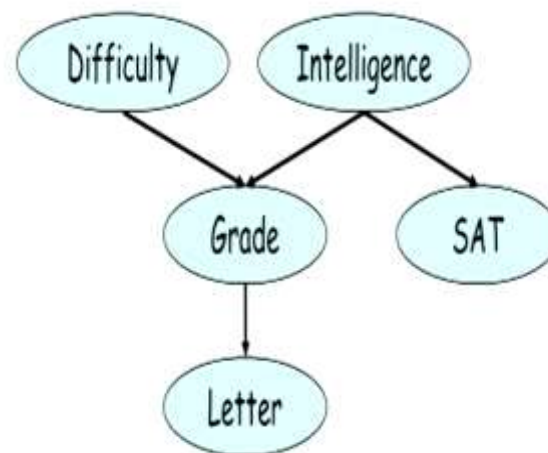
$$X \leftarrow Y$$

$$X \rightarrow W \rightarrow Y$$

$$X \leftarrow W \leftarrow Y$$

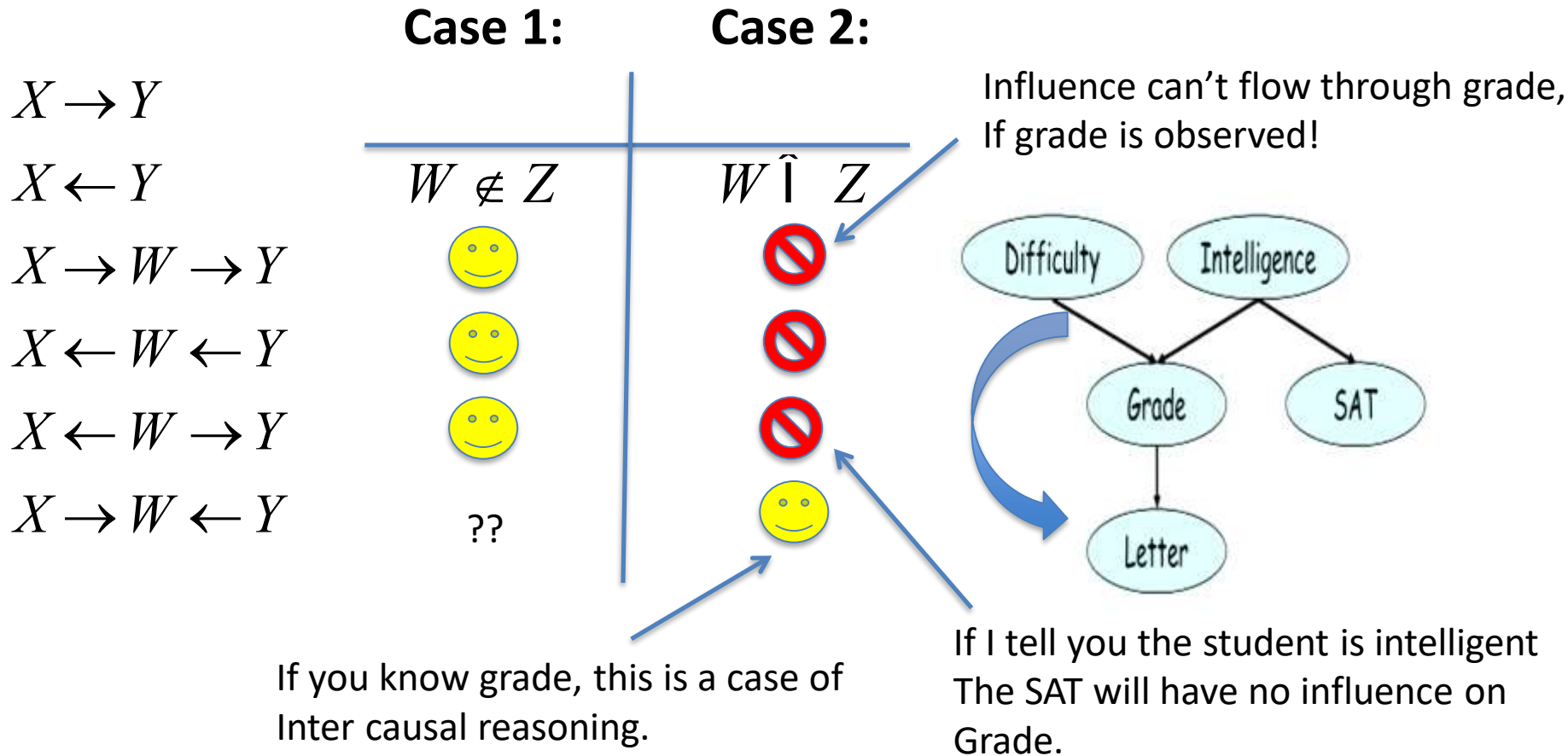
$$X \leftarrow W \rightarrow Y$$

$$X \rightarrow W \leftarrow Y$$



Bayesian Network

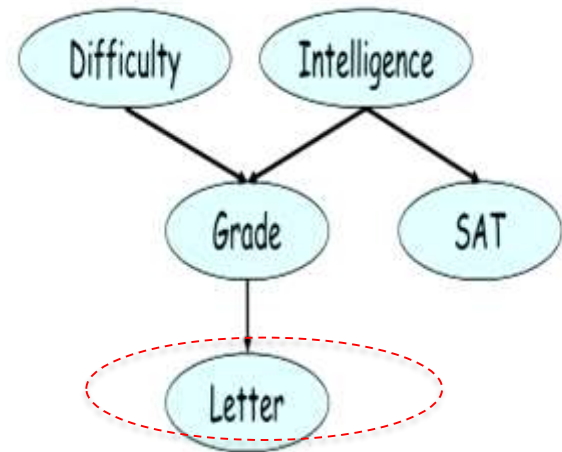
When can X influence Y given evidence about Z:



Bayesian Network

When can X influence Y given evidence about Z:

	Case 1:	Case 2:
	$W \notin Z$	$W \hat{=} Z$
$X \rightarrow Y$		
$X \leftarrow Y$		
$X \rightarrow W \rightarrow Y$	☺	⊘
$X \leftarrow W \leftarrow Y$	☺	⊘
$X \leftarrow W \rightarrow Y$	☺	⊘
$X \rightarrow W \leftarrow Y$	☺	☺



If you know don't observe grade directly, but you observe letter?



if W and all of its descendants are not observed.



if W or one of its descendants is observed.

Independencies in Bayesian Network

Recall Independence and factorizations:

- $P(X, Y) = P(X)P(Y)$ if X and Y are independent
- $P(X, Y, Z) = f_1(X, Z)f_2(Y, Z)$ if $X \perp Y | Z$, X and Y are conditionally independent given a third variable Z .

Factorization of a distribution P implies independencies that hold in P .

- Key point which enables us to factor graphs/ and joint distributions in high dimensional space.
- In particular, **P factorizes over G** , and we can read the independencies directly from the structure.

Flow of Inference and d-separation

Definition: X and Y are d-separated in G given Z if there is no active trail in G between X and Y given Z

Notation:

$$\text{d-sep}_G(X, Y | Z)$$

Flow of Inference and d-separation

Theorem: If P factorizes over G , and $d\text{-sep}_G(X, Y \mid Z)$, then P can be represented as the compact product of conditional probabilities.

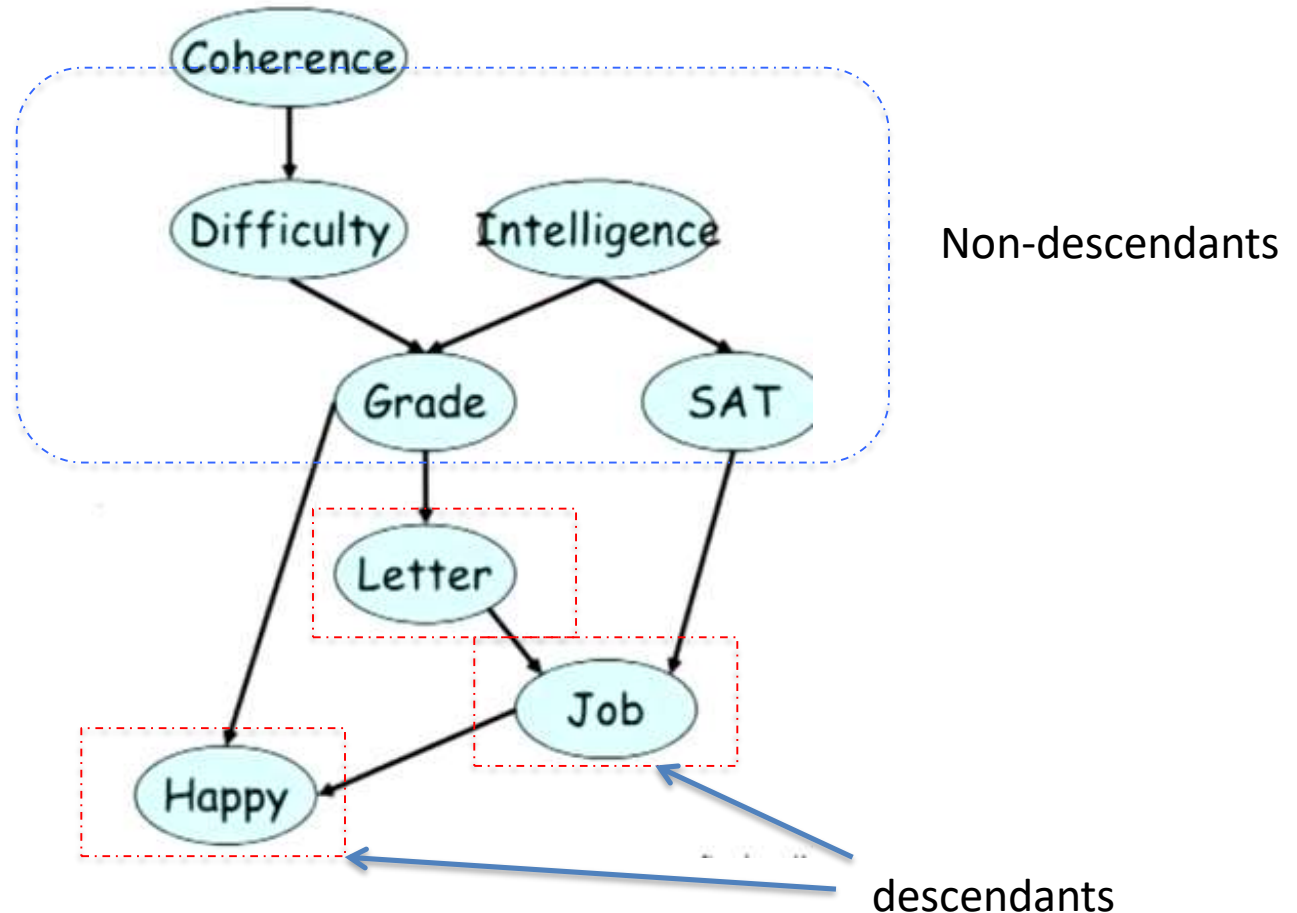
(D-separation implies independence if the distribution factors over the graph).

Example:

$$P(D, I, G, S, L) = P(D)P(I)P(G \mid D, I)P(S \mid I)P(L \mid G)$$

Flow of Inference and d-separation

Property: Any node is d-separated from its non-descendants given its parents.



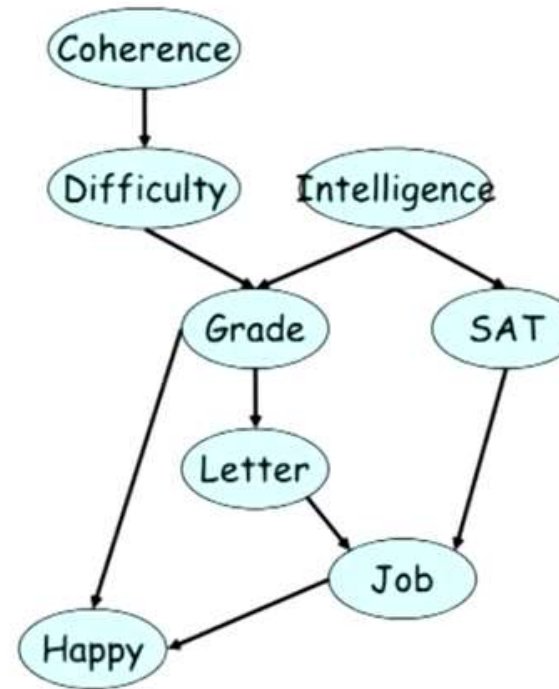
Flow of Inference and d-separation

A test:

Lets look at Letter and SAT.

Can we find a trail that is active given the parents (grade)?

NO!



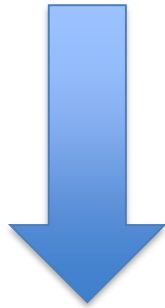
Flow of Inference and d-separation

A test:

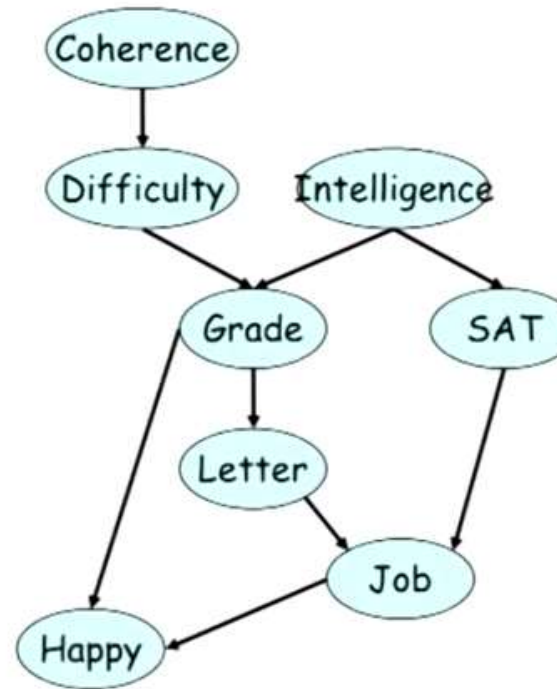
Lets look at Letter and SAT.

Can we find a trail that is active given the parents (grade)?

NO!



If P factorizes over G , then in P , any variable is independent of its non-descendants given its parents.



I-map

- D-separation in $G \rightarrow P$ satisfies the corresponding independence statement:

$$I(G) = \{ X \perp Y \mid Z : \text{d-sep}_G(X, Y \mid Z) \}$$

Set of all independencies in a graph G

All of the independencies derived from d-separation

*This set of independencies $I(G)$ is called an “I-map”.

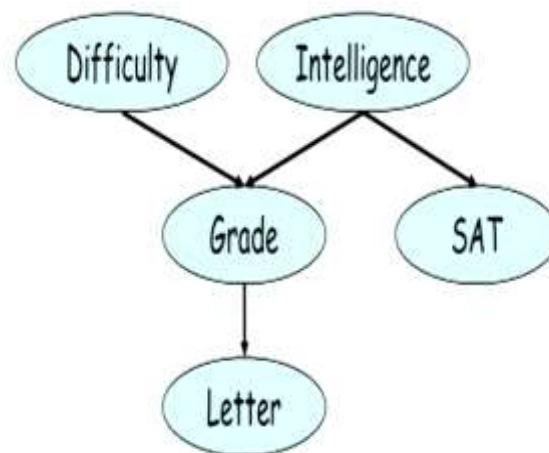
More formally: We say if P satisfies $I(G)$, then G is an I-map (independency map) of P .

I-map

A closer look (the converse also holds):

Theorem: If G is an I-map for P , then P factorizes over G .

$$P(D, I, G, S, L) = P(D)P(I | D)P(G | D, I)P(S | D, I, G)P(L | D, I, G, S)$$



I-map

A closer look (the converse also holds):

Theorem: If G is an I-map for P , then P factorizes over G .

$$P(D, I, G, S, L) = P(D)P(I | D)P(G | D, I)P(S | D, I, G)P(L | D, I, G, S)$$

$P(I)$

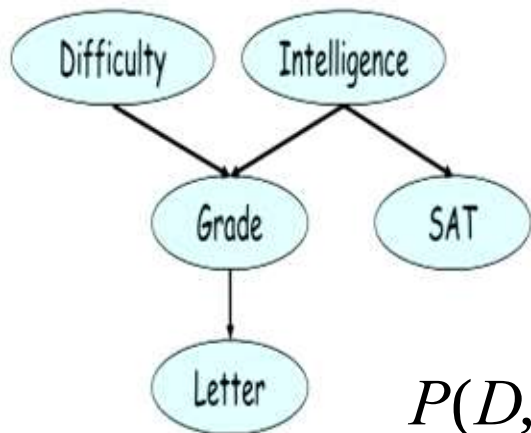
I and D are independent

$P(S | I)$

S is independent of D and G
Given I .

$P(L | G)$

L is independent its
non-descendants given its
parents



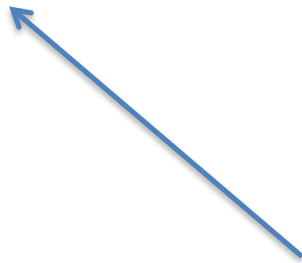
$$P(D, I, G, S, L) = P(D)P(I | D)P(G | D, I)P(S | I)P(L | G)$$

I-map / Independence Summary

We have established two equivalent views of a graph structure:

- Factorization: G allows P to be represented.
- I-map: Independencies encoded by G to hold in P .

If P factorizes over a graph G , we can read from the graph the **independencies** that must hold in P (independencies map).



A big deal

What depends on what

What does not depend on what

What happens when you change something

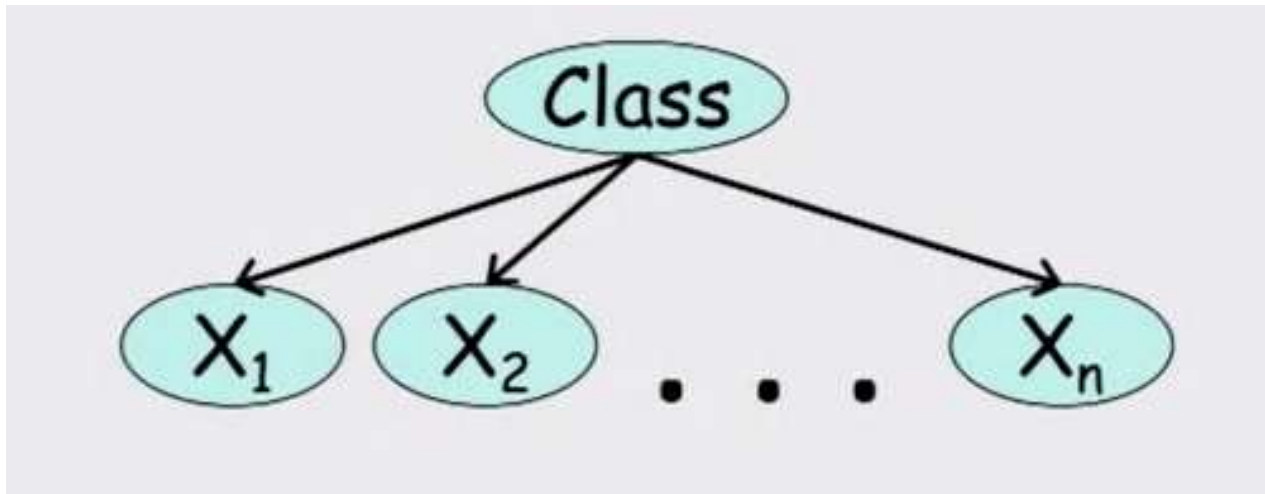
Consequences upstream and downstream

Naïve Bayes (idiots Bayes')

- Subclass of Bayesian Networks – make independence assumptions that are often unrealistic and overly simplistic.
- Often can be fairly accurate.
- Tradeoff in model complexity
- Typically used for classification.

Hidden Class
Features
(Infer)

Observed
Features
(Data)

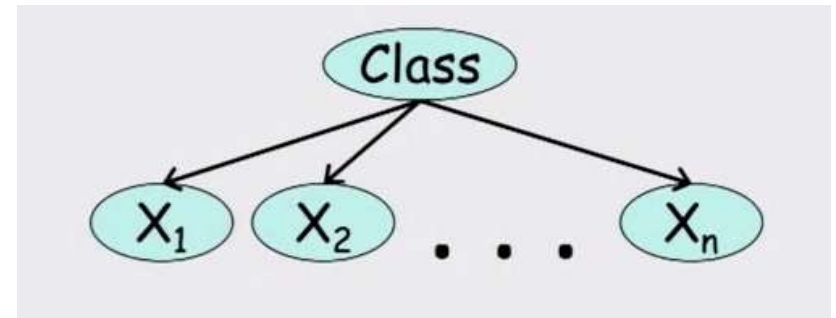


What is the independence assumption here?

Naïve Bayes Classifier

- $P(X_i \wedge X_j | C)$ for all X_i, X_j .

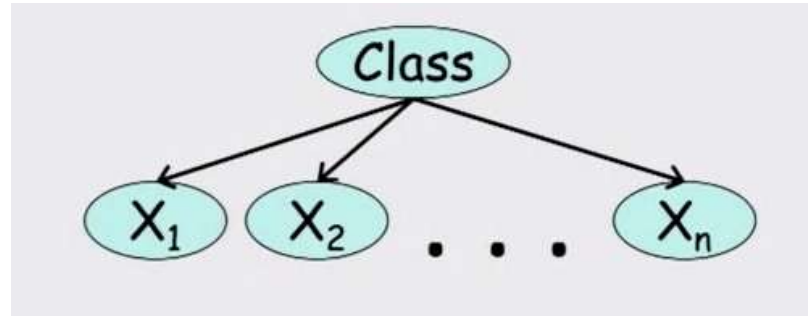
In other words, once class is observed, there is no additional information to observe from taking into account additional variables.



- The joint distribution is straightforward:

$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i | C).$$

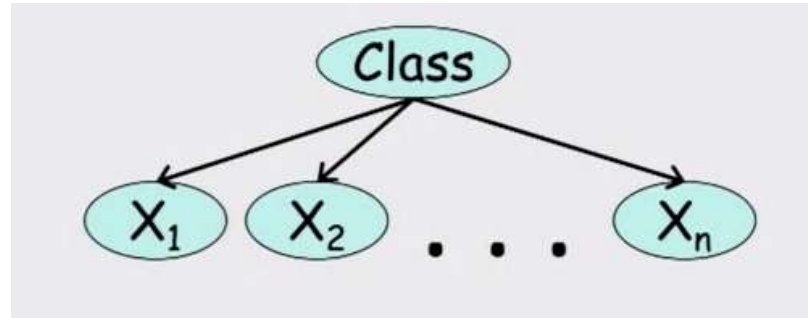
Naïve Bayes Classifier



- Of value to consider the ratio:

$$\frac{P(C = c^1 \mid x_1, \square, x_n)}{P(C = c^2 \mid x_1, \square, x_n)}$$

Naïve Bayes Classifier



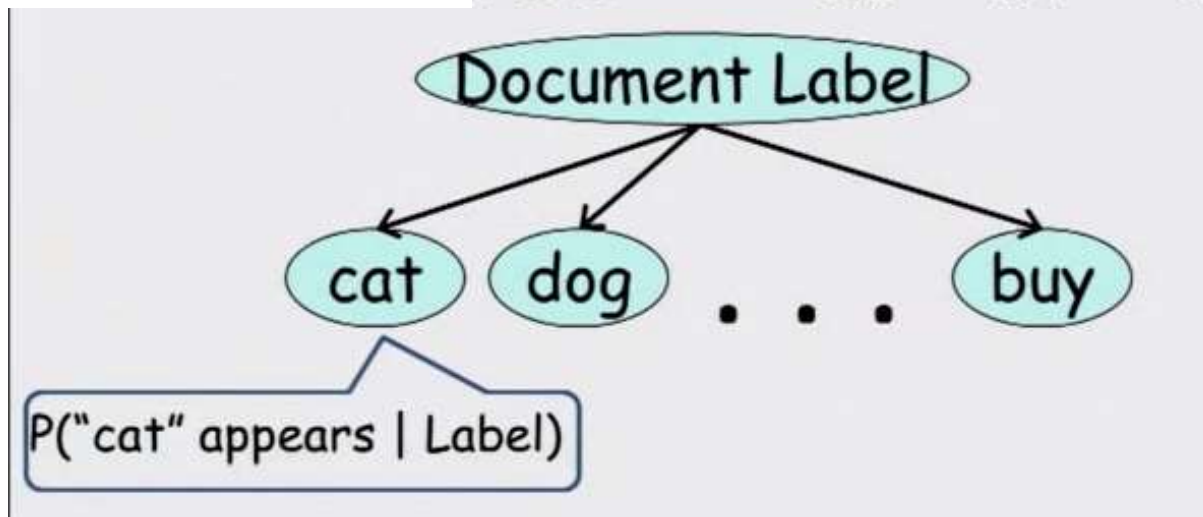
- Of value to consider the ratio:

$$\frac{P(C = c^1 | x_1, \dots, x_n)}{P(C = c^2 | x_1, \dots, x_n)} = \frac{P(C = c^1)}{P(C = c^2)} \prod_{i=1}^n \frac{P(x_i | C = c^1)}{P(x_i | C = c^2)}$$

Naïve Bayes Classifier

- Bernoulli Naïve Bayes for Text Classification:
treats every word in the dictionary as a binary rv.

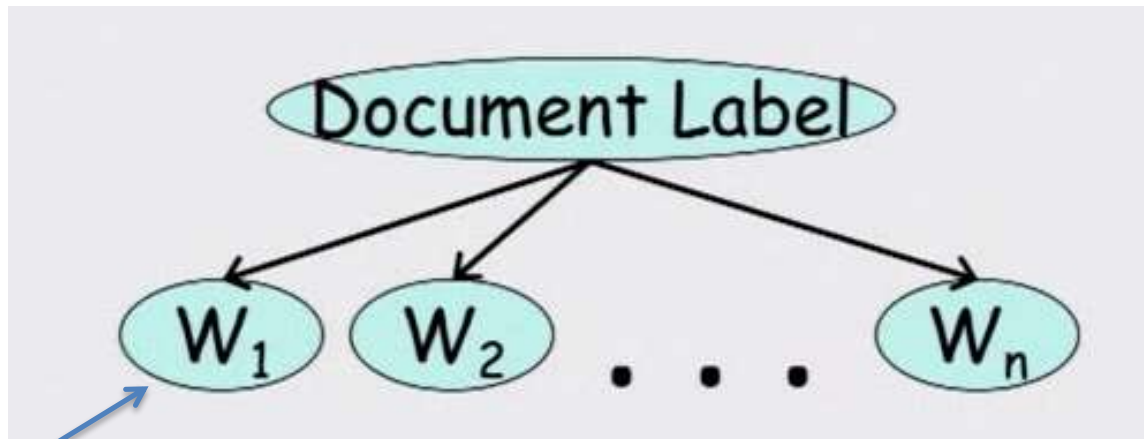
	cat	dog	buy	sell	...
Financial	0.001	0.001	0.2	0.3	...
Pets	0.3	0.4	0.02	0.0001	...



$$\frac{P(C = c^1 \mid x_1, \dots, x_n)}{P(C = c^2 \mid x_1, \dots, x_n)} = \frac{P(C = c^1)}{P(C = c^2)} \prod_{i=1}^n \frac{P(x_i \mid C = c^1)}{P(x_i \mid C = c^2)}$$

Naïve Bayes Classifier

- Multinomial Naïve Bayes for Text Classification:
treats every variable as a word in the document. The value, is the actual word, and there are n words in the document.



An actual first word from the document

$$\frac{P(C = c^1 | x_1, \square, x_n)}{P(C = c^2 | x_1, \square, x_n)} = \frac{P(C = c^1)}{P(C = c^2)} \prod_{i=1}^n \frac{P(x_i | C = c^1)}{P(x_i | C = c^2)}$$

Naïve Bayes Classifier

A Comparison of Event Models for Naive Bayes Text Classification

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klaR: Classification and visualization

Miscellaneous functions for classification and visualization developed at the Fakultät Statistik, Technische Universität Dortmund

Version: 0.6-10
Depends: R (≥ 2.0.0), [MASS](#)
Imports: [combinat](#)
Suggests: [scatterplot3d](#) (≥ 0.3-22), [som](#), [mlbench](#), [rpart](#), [e1071](#), [evtree](#)
Published: 2014-02-11

e1071: Misc Functions of the Department of Statistics (e1071), TU Wien

Functions for latent class analysis, short time Fourier transform, fuzzy clustering, support vector machines, shortest path computation, bagged clustering, naive Bayes classifier, ...

Version: 1.6-2
Depends: [class](#)
Imports: graphics, stats, grDevices, utils
Suggests: [cluster](#), [mlbench](#), [nnet](#), [randomForest](#), [rpart](#), [SparseM](#), [xtable](#), [Matrix](#), [MASS](#)
Published: 2014-01-11

Naïve Bayes Summary

- Simple approach for classification:
 - Computationally efficient
 - Easy to construct
- Surprisingly effective in domains with many relatively weakly relevant features.
- Strong independence assumptions reduce performance when many features are correlated.

Application: Medical Diagnosis

Medical Diagnosis

PhD thesis of Heckerman et al. (1992) Pathfinder.

- Range of information to help doctors diagnose disease.
- Tried a bunch of different methods.
- Help pathologist diagnose lymph node pathologies for 60 different diseases.

Application: Medical Diagnosis

Medical Diagnosis

- Pathfinder I: Rule-based system.
- Pathfinder II: Naïve Bayes superior performance.
- Pathfinder III: Naïve Bayes with better knowledge engineering.
 - do not put in zeros, unless its really zero!
 - Better calibration with physicians. Physicians can better distinguish characteristics between diseases compared to within disease.
- Pathfinder IV: Full Bayesian network
 - removed incorrect independencies
 - Additional parents led to more accurate estimates of probabilities.

Application: Medical Diagnosis

BN model agreed with **expert panel** in 50/53 cases, vs. 47/53 for the naïve Bayes model, significantly less for the rule based system. (The hard cases).

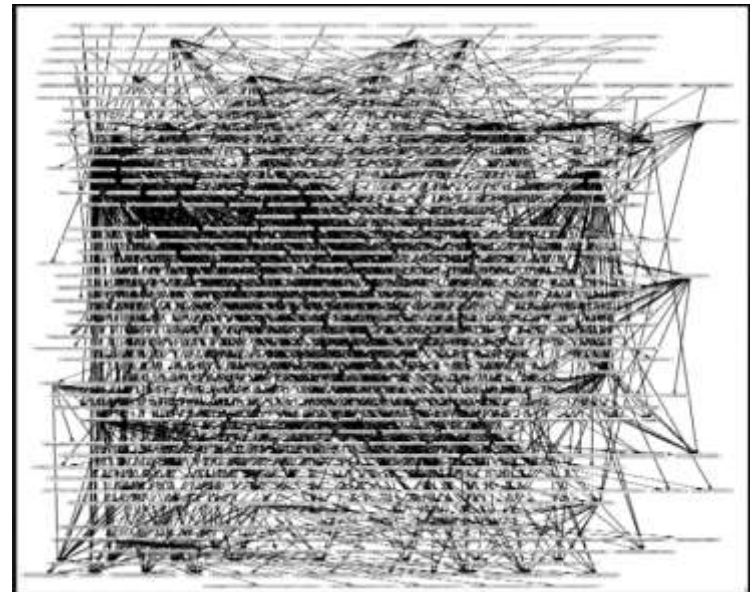
- It outperformed the panel of physicians that designed the model.
Why?

We cannot process the interaction of so many factors in our heads? We can't put together all of these findings in a systematic way.

Application: Medical Diagnosis

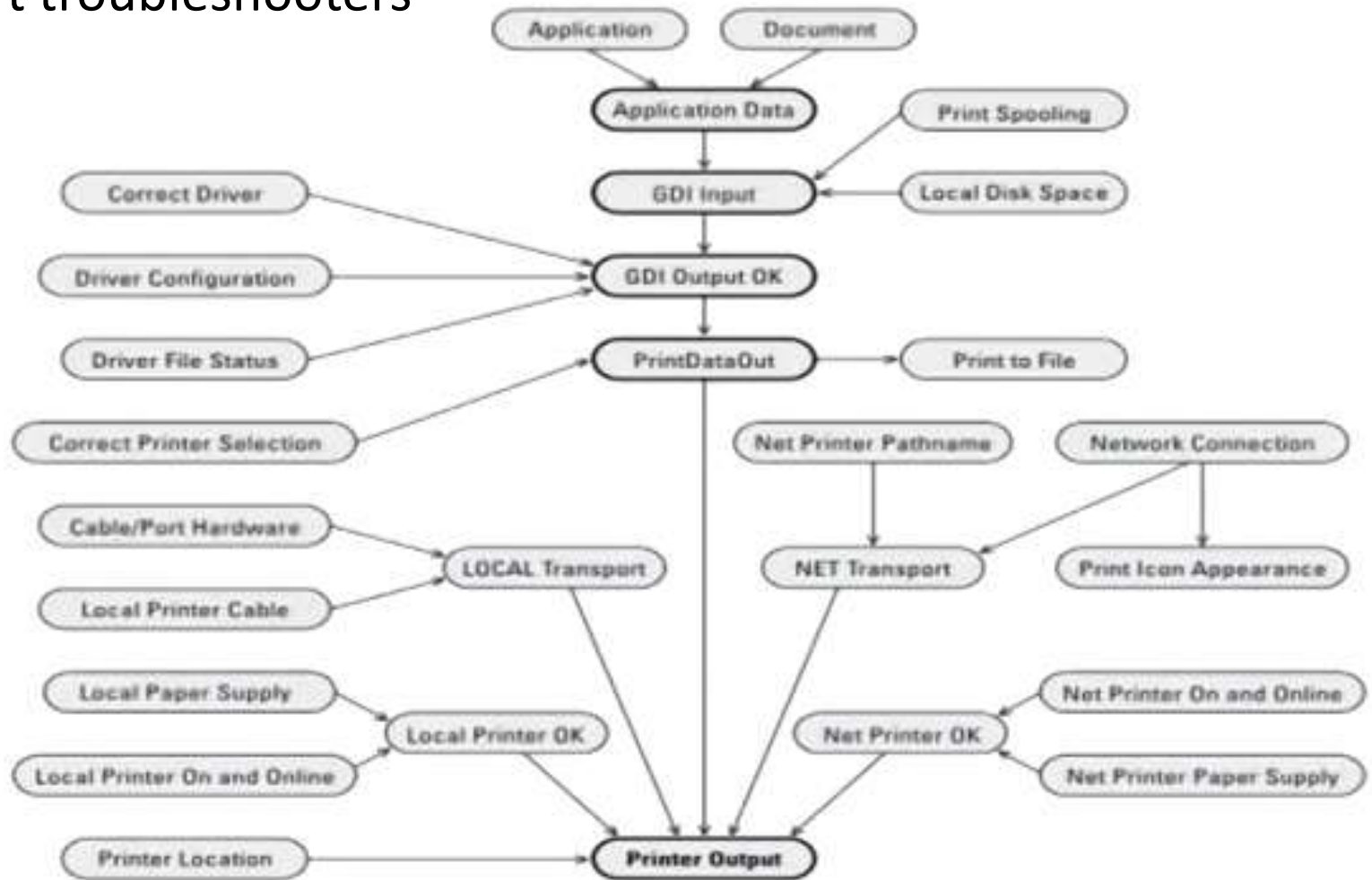
Another example: Computer-based Patient Case Study (CPCS network).

- ~ 500 variables, each with ~ 4 values.
- $\sim 4^{500}$ variables, to specify the complete joint distribution.
- Simplification on CPD to make the problem tractable.



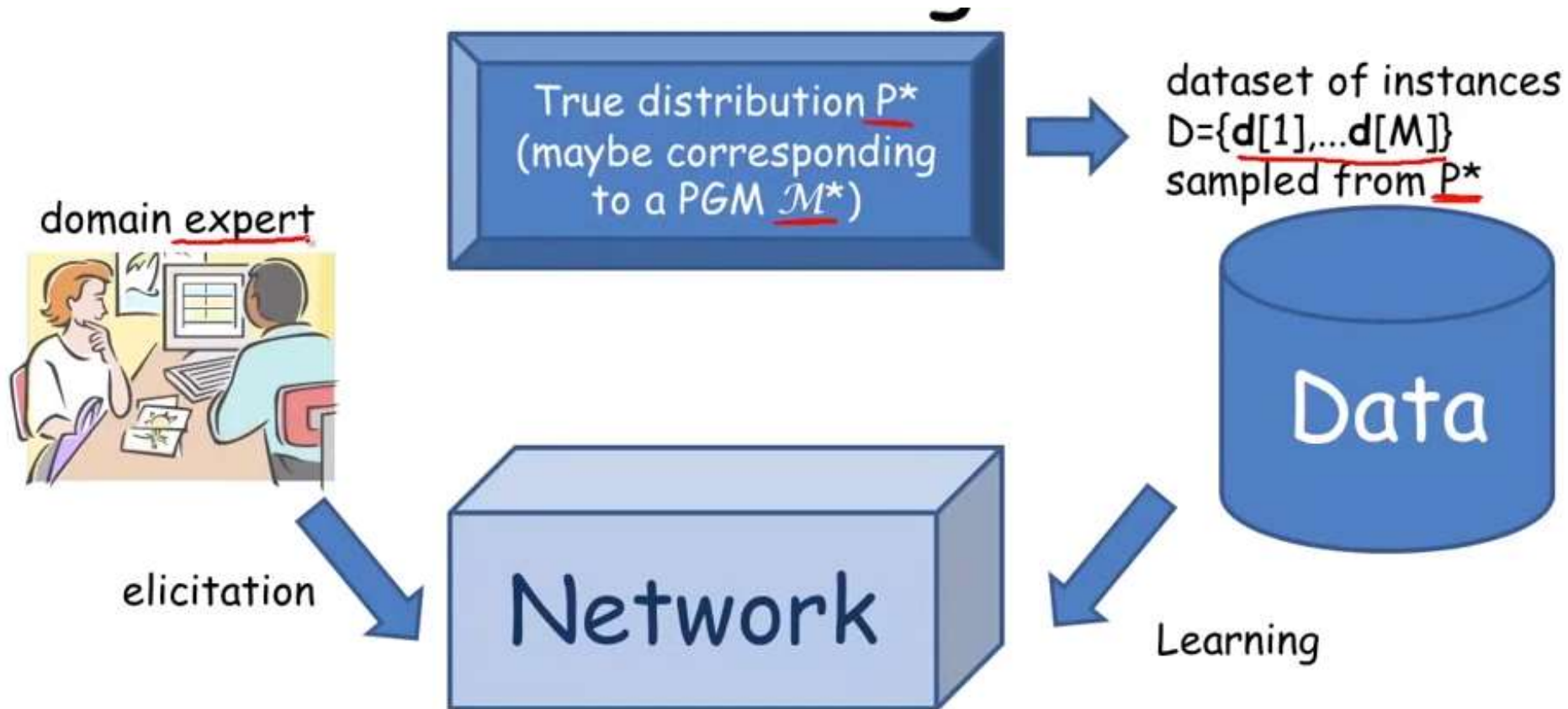
Application: Fault Diagnosis

Microsoft troubleshooters



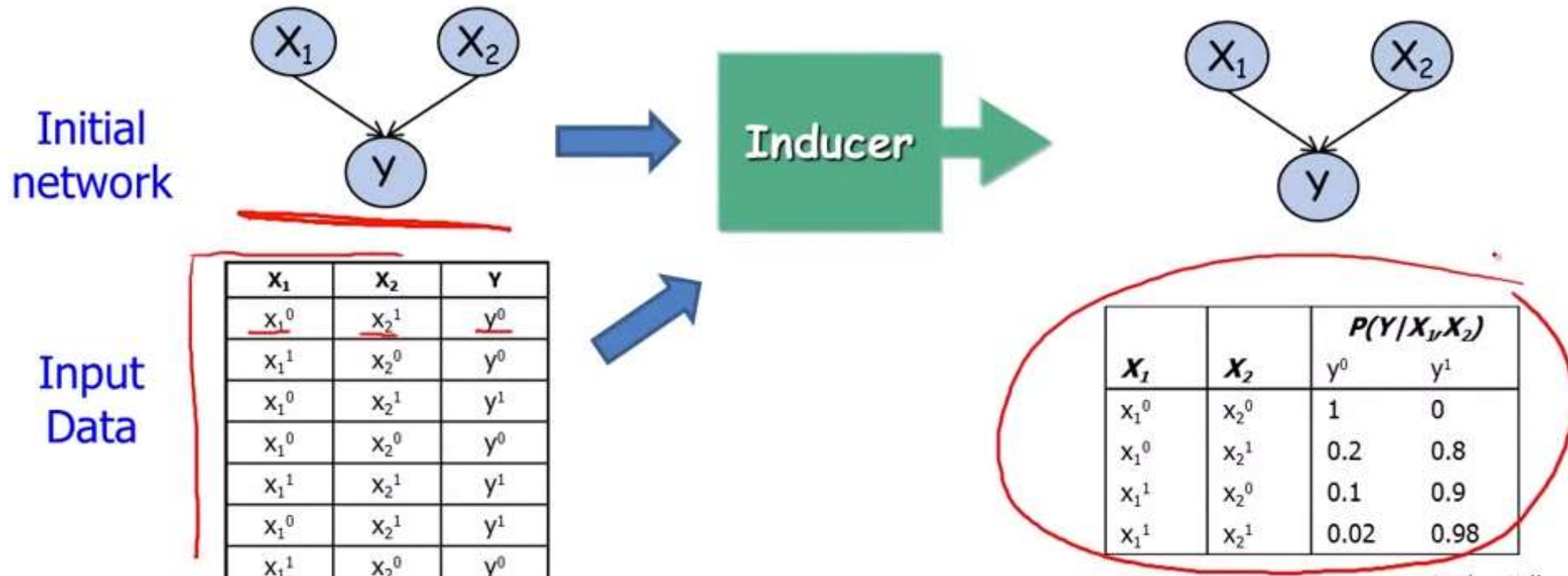
Fault diagnosis also in car repair

PGM learning from data



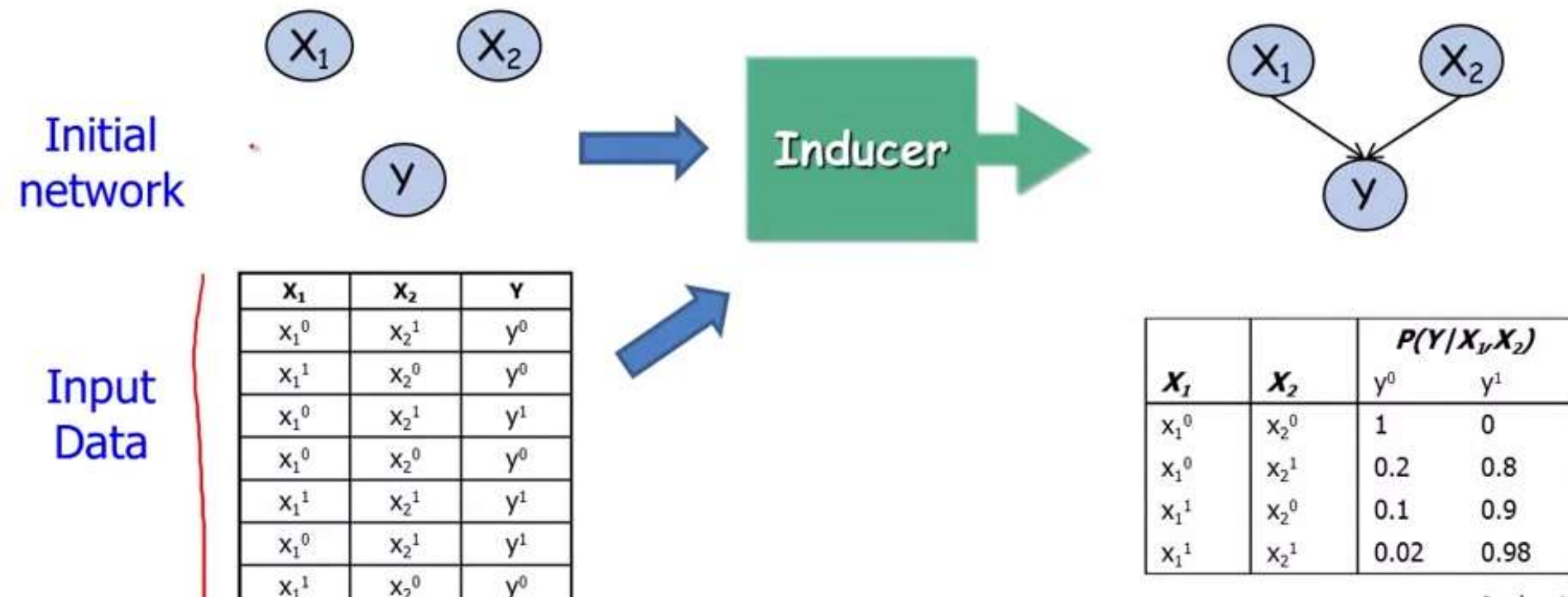
PGM learning from data

Known Structure, Complete Data



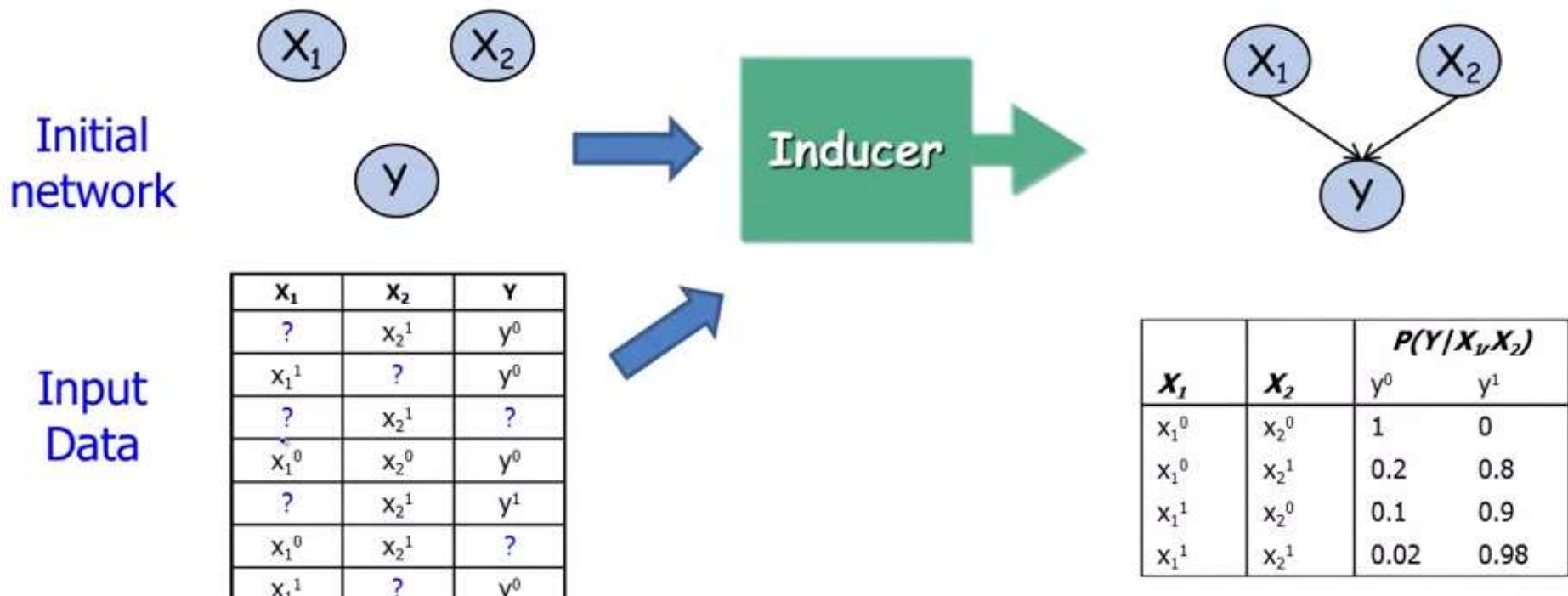
PGM learning from data

Unknown Structure, Complete Data



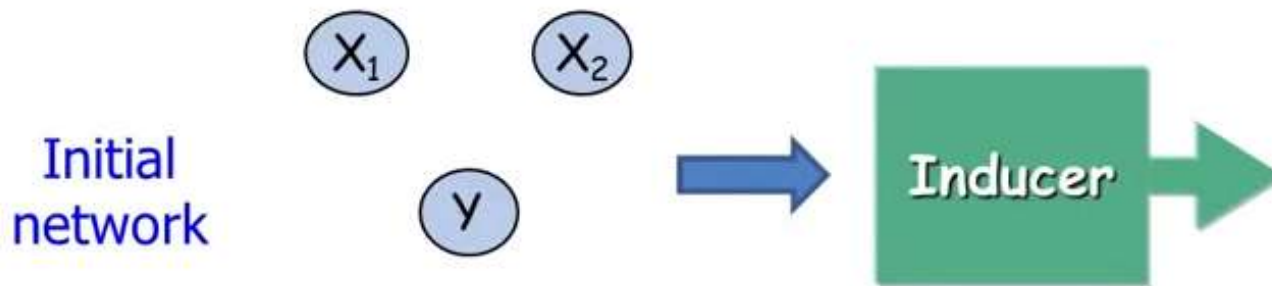
PGM learning from data

Unknown Structure, Incomplete Data



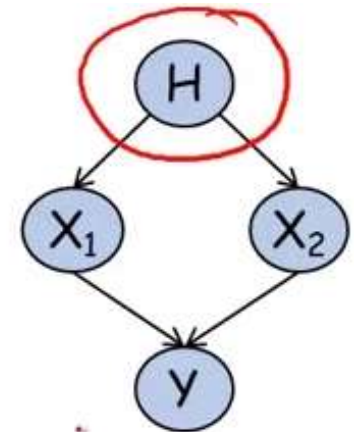
PGM learning from data

Unknown Structure, Latent Variables, Incomplete Data



Input Data

x_1	x_2	y
?	x_2^1	y^0
x_1^1	?	y^0
?	x_2^1	?
x_1^0	x_2^0	y^0
?	x_2^1	y^1
x_1^0	x_2^1	?
x_1^1	?	y^0



x_1	x_2	$P(Y x_1, x_2)$	
		y^0	y^1
x_1^0	x_2^0	1	0
x_1^0	x_2^1	0.2	0.8
x_1^1	x_2^0	0.1	0.9
x_1^1	x_2^1	0.02	0.98

PGM learning from data: Task 1

- **Goal:** Answer general probabilistic queries about new instances.
- **Simple Metric:** Training set likelihood

$$P(D:M) = \prod_m P(d[m]:M).$$

data model Instances over
a model

How probable are the instances we observe
in a given model?

- Seems like a good idea, but we really care about new data.
 - Ideally, evaluate on a test set likelihood: $P(D':M)$
will be better in terms of generalization performance.

PGM learning from data: Task 2

- **Goal:** Specific prediction task on new instances
- Predict target variables “Y” from observations from “X”
e.g., image segmentation.
- Often care about specialized objective
e.g., pixel level segmentation accuracy.
- Often convenient to select a model to optimize:
 - likelihood $\tilde{\mathbf{O}}_m P(d[m]:M)$.
 - Conditional likelihood $\tilde{\mathbf{O}}_m P(y[m] | x[m]:M)$.

Model should be thrown at test data!

PGM learning from data: Task 3

- **Goal:** Knowledge discovery of M^* :
 - Distinguish direct vs. indirect dependencies.
 - Possibly directionality of edges.
 - Presence and location of hidden variables.
- Often train using likelihood
 - Poor surrogate for structural accuracy.
- Evaluate by comparing to prior knowledge or “gold standard”.

Avoid Overfitting

- Selecting M^* to optimize the training set likelihood will overfit parameters.
 - Can combat with parameter priors.
- Structure overfitting
 - Complex structures always win – training likelihood decreases as model complexity increases.
 - Must penalize or bound model complexity.

Selecting Hyper-parameters

- Regularization for overfitting involves hyperparameters
 - parameter priors
 - complexity parameters
 - regularization
- Choice of hyper-parameters makes a HUGE difference, but non-trivial to select.

Why PGM Learning

- Often in a position when predictions of structured objects, e.g., sequence, graphs, trees.
- Exploit correlations between variables.
- Can incorporate prior knowledge easily into the algorithm.
- Learning a “single model” which describes multiple tasks.
- Framework for knowledge discovery.